

infectious diseases

Infectious disease: a disease that is caused by a pathogen and is transmissible (can spread)

Infectious diseases you need to know:

- (i) Cholera
- (ii) Malaria
- (iii) HIV/AIDS
- (iv) Tuberculosis (TB)

Cholera

① What is the causative pathogen?

caused by a bacterium, Vibrio cholerae

② How does the disease spread?

fecal matter of an infected person, contaminates food & water that is ingested by another person



③ What are the symptoms?

* diarrhea (→ leads to loss of water & salts from the body) & vomiting

④ How do we treat the disease?

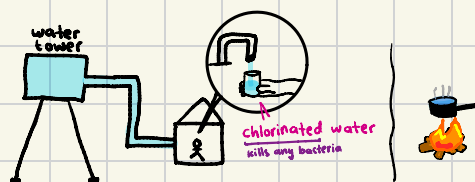
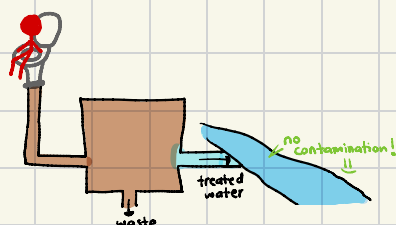
- lost a lot of water & salts
- V. cholerae present in their gut
- fluid & salt replacement therapy
- antibiotics (details later)
- drinking oral rehydration salts
- intravenous therapy (injecting water & salts in person)

⑤ How do we prevent the disease?

- sewage treatment facilities

- providing clean, piped water or
boil & cool water before drinking

- eat food hot & cover it



⑥ Why can't we eradicate the disease?

- sewage treatment is expensive
(difficult to have for poorer countries)

- natural disasters can destroy
sewage water facilities & pipe lines

- tendency to eat raw shellfish
& water crop plants with contaminated water

↓
faeces will accumulate

↓
no access to clean water

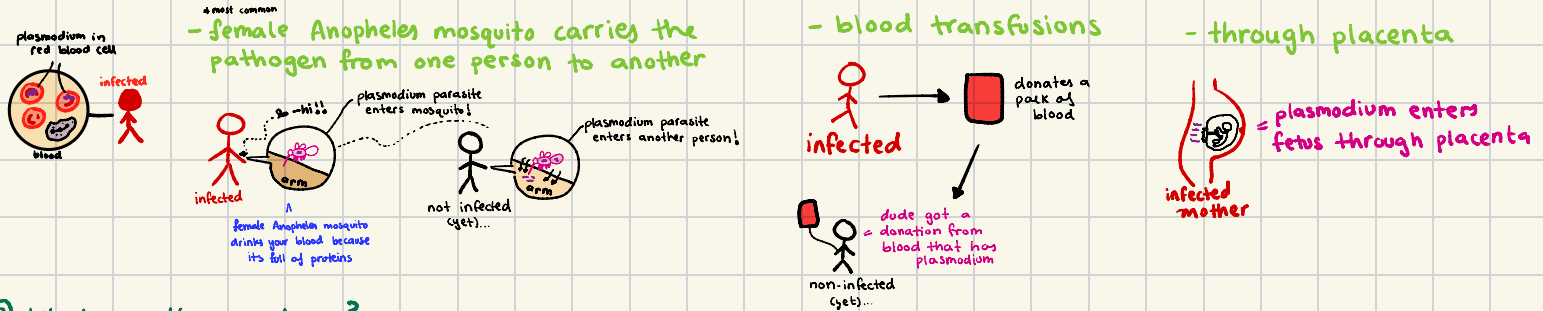


Malaria

① What is the causative pathogen?

caused by a protist, Plasmodium malariae

② How does the disease spread?



③ What are the symptoms?

* do not need to know [fever, nausea, anaemia, headaches, muscle pain]

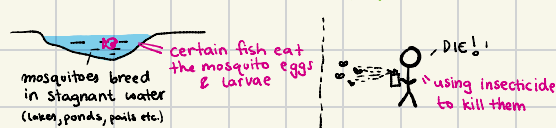
④ How do we treat the disease?

* do not need to know in detail

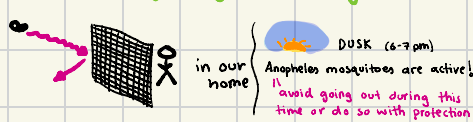
JUST KNOW: * antimalaria drugs (these drugs kill the plasmodium parasite) e.g. chloroquine

⑤ How do we prevent the disease?

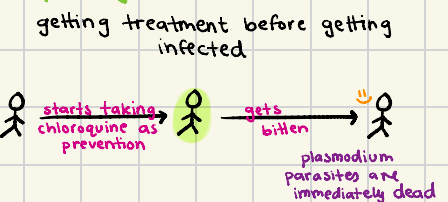
- reduce the mosquito population



- prevent getting bitten by mosquitoes



- prophylaxis



⑥ Why can't we eradicate the disease?

- some plasmodium parasites evolve to become drug resistant or insecticide resistant (antimalarial drugs are less effective)

prophylaxis won't work

- mosquitoes easily breed & reproduce in hot, tropical regions (especially in rainy seasons)

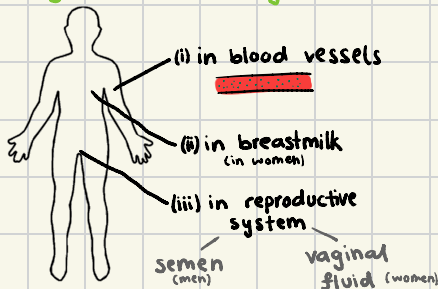
HIV/AIDS (HIV leads to AIDS IF very serious)

① What is the causative pathogen?

caused by the virus, human immunodeficiency virus

② How does the disease spread?

by various bodily fluids:



- blood transfusion



- breastfeeding



- transfer of contaminated semen or vaginal fluid
⇒ through unprotected sex (oral, vaginal, anal)

- contaminated needles



- through placenta



can also happen during birth
baby passes through vagina & could come into contact with contaminated vaginal fluid

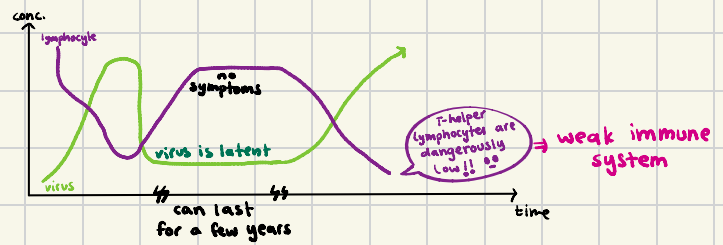
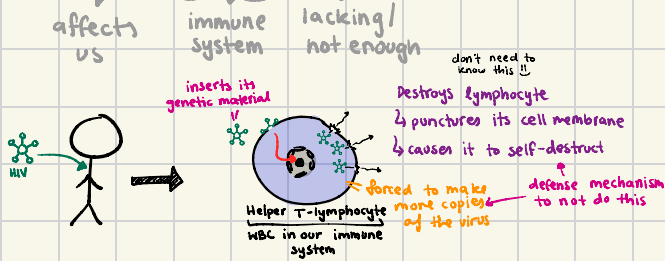
② What are the symptoms? * takes 2-6 months to infect a person

→ no symptoms for most people or short term flu-like symptoms (fever, body ache)

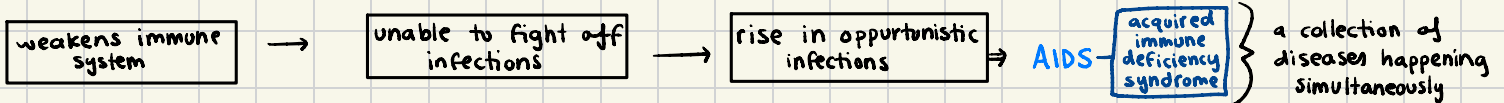
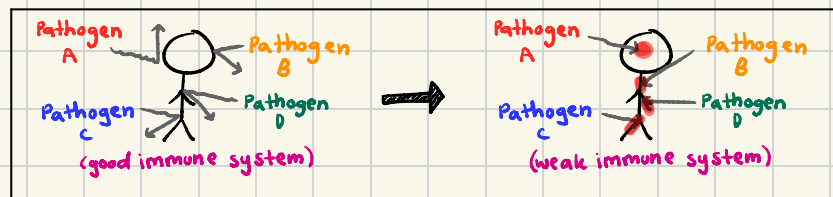
so why is it serious??

human immunodeficiency virus

⇒ it's a virus that makes our immune system weaker



HIV weakens your immune system & thus your body is also unable to fight off other pathogens.



④ How do we treat the disease?

no cure yet BUT there is Antiretroviral Therapy (ART) → keeps the virus latent (inactive)

is made up of a combination of drugs → prevents the viral genetic material from entering the nucleus of the T-cells

needs to be taken daily
for it to be effective

⑤ How do we prevent the disease?

- test the blood that is donated
- giving drug users clean, sterile needles (needle exchange programs)
- no unprotected sex (use condoms, femidoms, dental dams (prevents contact with semen/vaginal fluid
- contact tracing (identifying people with HIV to inform them of their risks in spreading the disease)
- start HIV+ people on ART immediately (cuts chances of transmission to almost 0%) ← [also for pregnant/breastfeeding women]

⑥ Why can't we eradicate the disease?

⇒ limitations of treatment

- ↳ ART is not a cure & HAS to be taken daily to be effective & can have undesirable side effects
- ↳ ART is expensive in poorer countries

⇒ limitations of prevention

- ↳ condoms, femidoms & dental dams are not 100% effective
- ↳ feeling shameful in getting tested 😞

Tuberculosis (TB)

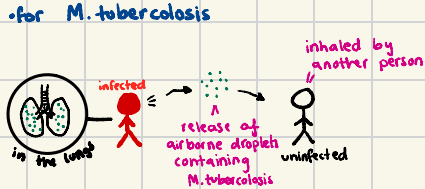
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① What is the causative pathogen?

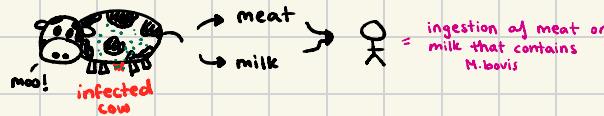
Mycobacterium tuberculosis & Mycobacterium bovis

② How does the disease spread?

• for *M. tuberculosis*



• for *M. bovis*

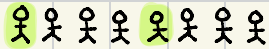


③ What are the symptoms?

* most cases affect the lungs → shortness of breath, weight loss

LATENT TB

≈ 20% of the world pop. has latent TB



⇒ When does the latent TB become active?

* malnourishment, diabetes, smoking (damages lungs)

* a weakened immune system due to HIV!

④ How do we treat the disease?

★ Antibiotics - drugs/medicine that is able to kill bacteria or inhibit bacterial growth

⇒ multiple different antibiotics are given daily to patients with active TB

↳ this is because some *Mycobacterium* might be drug resistant

* antibiotics are given through DOTS [healthcare worker/family ensures patient takes the medication]

⑤ How do we prevent the disease?

→ quarantine of people with active TB (esp. the first 2-4 weeks)

→ contact tracing - identifying close contacts for suspicion of TB as well

→ vaccination

→ killing infected cattle & pasteurization of milk

→ improve nutrition of the person & improve housing conditions

⑥ Why can't we eradicate the disease?

- spreads easily through the air - causing many people to have latent TB

- many parts of the world suffer from malnourishment

- rise in people with HIV

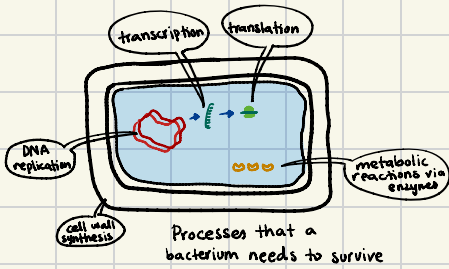
* *Mycobacterium* are evolving to become antibiotic resistant

Antibiotics

* drug / chemical

* able to kill bacteria or inhibit its growth

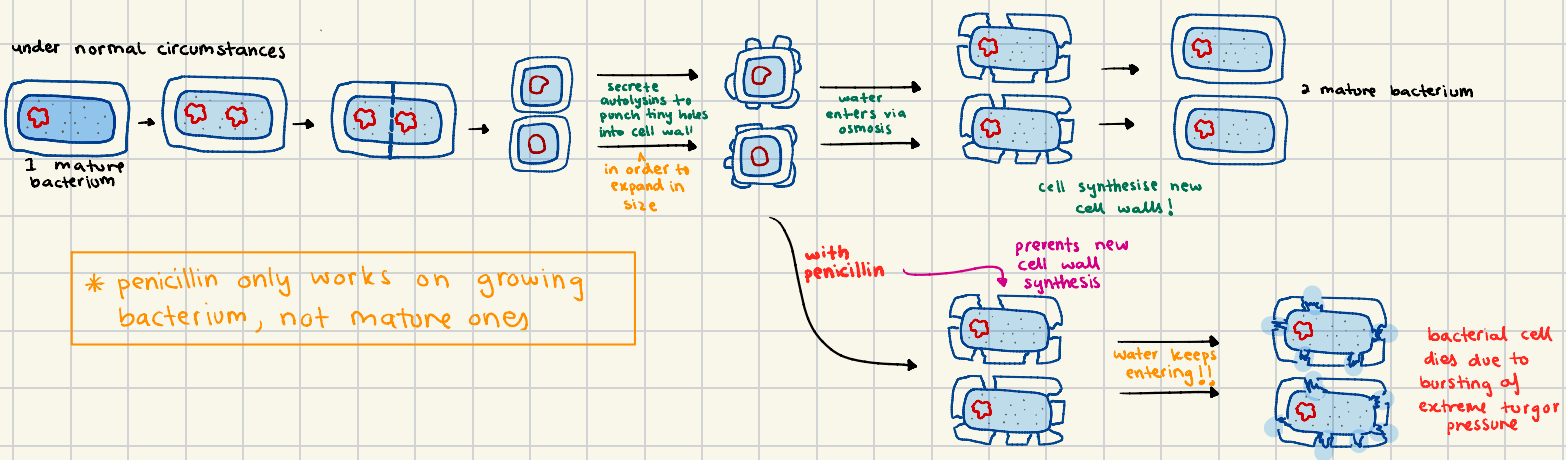
⇒ How do they work??



SPECIFIC ANTIBIOTICS BLOCK SPECIFIC PROCESSES OF THE BACTERIUM, CAUSING IT TO DIE!

e.g penicillin blocks synthesis of bacterium cell walls

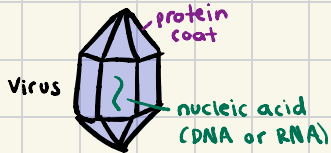
many different kinds - we only need to know about penicillin



* penicillin only works on growing bacterium, not mature ones

! antibiotics cannot affect viruses!

↳ why?



- ① they don't have a peptidoglycan cell wall
- ② viruses do not carry out transcription & translation
- ③ they don't carry out metabolic reactions by themselves

viruses lack target sites that are affected by antibiotics

Antibiotic resistance

↳ when antibiotics are no longer effective against bacteria

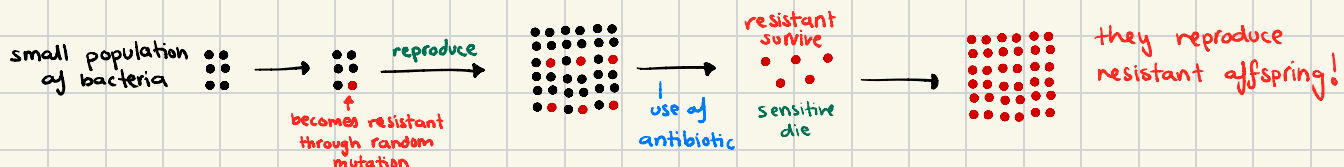
⇒ how does a bacterium become antibiotic resistant

① acquiring an antibiotic gene through mutation

② plasmid transfers (a resistant bacteria can transfer its antibiotic resistant gene to a non-resistant bacterium)

⇒ how does an entire population become antibiotic resistant

NATURAL SELECTION



Impact of antibiotic resistance

- ① if infections are not treated, they get more dangerous
- ② infected people can infect others

Reducing the impact of the resistance

- ① using DOTS to ensure the patients finish their antibiotics correctly
- ② using 2-3 different antibiotics simultaneously (increases chances of killing resistant bacteria)
- ③ avoid using antibiotics for viral infections
- ④ only use antibiotics when it is essential